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5100-02
Foundations of Data Science

Education: Communicate the Results

The purpose of this project was to address the significance of socioeconomic factors in academic performance in American High schools. Academic performance was measured with ACT scores and the socioeconomic factors evaluated were the median household income, participation in the free/reduced lunch programs, the rate of unemployment, the percent of students with married parents, and the percent of students who eventually went to college. The significance of these socioeconomic factors was determined by how strongly they were correlated with ACT scores and if they were strong candidates for predicting ACT scores.

There were two data sets used in this project. The first data set, Edgap data, was collected by the by the Memphis Teacher Residency program. The EdGap data contains the Average ACT scores and socioeconomic characteristics previously described, by school district from 2016 – 2017. This data is largely collected from the Census Bureau's American Community Survey and each state's respective department of education. The data was collected from 20 states, mostly in the East and mid-West of the United States of America. The second data set, school-info, is from the National Center for Education Statistics and contains the corresponding school information. This included school grade level, states local identification numbers, state agency status, the schools name and state.

An initial data assessment was done of both data frames to ensure they were proper candidates to address the question. Some limitations included that the Edgap data was considerably smaller than the school-info data and missing values. However, both are tidy or long format. This project used regression analysis to examine the relationship of ACT scores and the individual socioeconomic factors. An initial pair plot was created to glimpse at the pairwise relationships between variables and their histograms. The initial plot indicated potential relationships between the socioeconomic variables and ACT scores. However, the plot did reveal that some of the values were out of range for an ACT score(1-36) and percent lunch values (0-1.00). However, the data was determined to be a good candidate for answering the question.

Selected columns from the school-info data were merged to the Edgap data. Keeping the year, zip-code, school type, school level, and charter school status columns to the relevant schools included in the Edgap data. This merged data set was reduced, refined, and re-organized to become the education_clean.csv. The missing values for rate_unemployment, percent_college, percent_married, median_income, and percent_lunch were imputed with an iterative imputer.

The initial exploratory analysis was conducted by creating a heatmap of the numeric correlation between the variables. The resulting graph (figure.1) suggests a strong negative correlation with participation in free/reduced lunch and average ACT scores. There is also a strong positive correlation between median household income and the percent of student who will eventually go to college. Furthermore, there is implications of median household income being negatively

correlated with participation in the free/reduced lunch program. This is unsurprising as the household income is directly related to participate in free/reduced lunch programs, whose eligibility involves the federal poverty line. Continued initial impression of the new data set resulted in another pair plot. The relationships appear to be linear and the distributions for unemployment rate, percent married, and median household income are skewed with many outliers. This is further confirmed with a boxplot. One important impression is that several of the predictors appear correlated with each other as well as with ACT scores.

Regression models were used to asses the socioeconomic values ability to be predictors of ACT scores. Based on the previous evaluation, linear regressions were chosen. First three singular regression models were tested and then two multiple linear regressions were tested.

The singular regression models included the median household income, the percent of student who went to college, and the percent of unemployed. The linear regression for median income resulted in an r-squared that was only 0.211, the mean-squared-error(MSE) was 2.228, and the mean-absolute-error(MAE) was 1.712. The median household income is not a significant predictor of ACT scores. However, based on the shape of the data for this variable (figure 2.), a quadratic regression model was also tested. The resulting r-squared was 0.219 and MAE was 1.69. There was negligible improvement in the r-squared values. The linear regression evaluating the percent of students who went to college indicated a slight positive correlation to average ACT scores. The r-squared was 0.210, the MAE was 1.17, and the coefficient was 6.97. The linear regression for the percent unemployed was a slight negative correlation to average ACT scores. The r-squared was 0.188, the MAE was 1.74, and the coefficient was -19.21. Both the percent college and percent unemployed variables in these regressions appear to be significant predictors on their own.

The multiple linear regression model evaluated all the socioeconomic factors as predictors for the average ACT scores. This resulted in an r-squared of 0.628 and a MAE of 1.14. The multiple linear regression model also returned p-values larger than 0.05 for percent of students with married parents (0.577) and the median household income (0.920). Thus, neither were significant predictors of ACT scores. Furthermore, their coefficients were notably small at -0.075 for percent-married and -1.21e-07 for median income. From this model, a reduced multiple linear regression model was made for the remaining three factors. The resulting r-squared was still 0.628 and MAE was still 1.14. The other meaningful portion of the reduced multiple factor regression was that it was clear that one socioeconomic factor was much stronger as a predictor for ACT scores than the others. The coefficient for unemployment was -2.269 and for percent college, it was -1.739. The coefficient for percent of students participating in free/reduced lunch was -7.805.

Further manipulation of the reduced multiple linear regression included normalizing the model. The mean would now be zero and the standard deviation was one. Doing this did not change the r-squared value or the MAE. However, it did scale down the coefficients. The

coefficient for unemployment rate was -0.122, for percent college it was 0.282, and for percent lunch it was -1.777. Still indicating a strong negative correlation for ACT scores and participation in the free/reduced lunch program.

The results of the education project indicate that some socioeconomic factors are significant in academic performance in American High schools. The socioeconomic factors that served as significant predictors for average ACT scores were the percent of student unemployment after High school, the percent who went to college after High school, and participation in a free/reduced lunch program. The strongest predictor being student participation in a free/reduced lunch program. The socioeconomic factors that were not significant predictors of average ACT scores were median household income and having married parents at home.

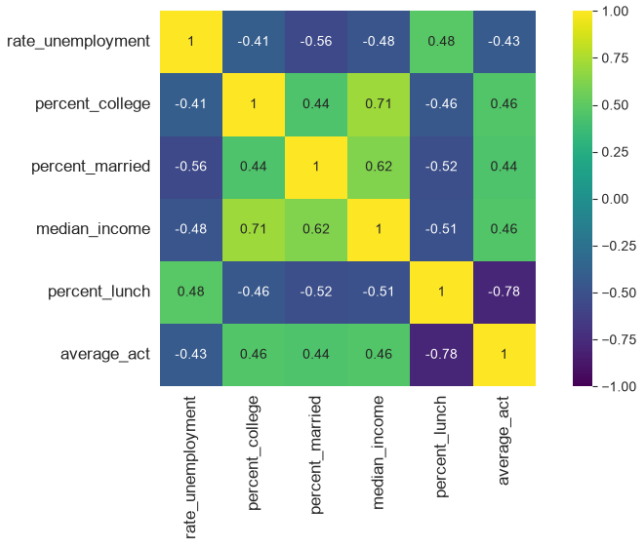


Figure 1

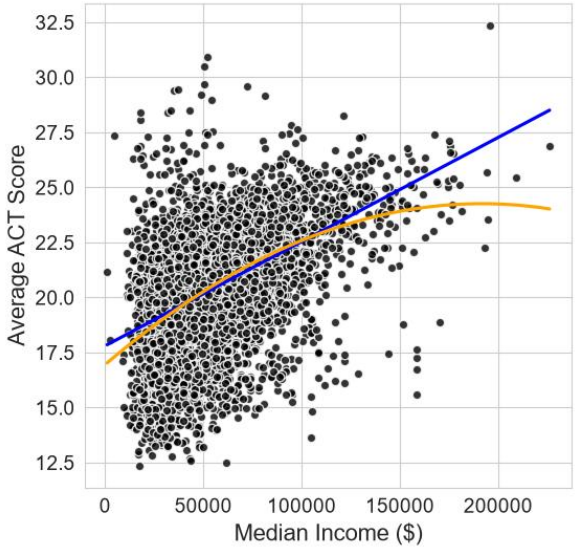


Figure.2